

Cauchy Distribution

①

A r.v 'x' has a cauchy distribution if the density function is of the form.

$$f(x) = \frac{1}{\pi [1^2 + (x - \mu)^2]} \quad -\infty < x < \infty$$

Q. what is standard cauchy distribution.

Sol

$$f(x) = \frac{1}{\pi [1^2 + (x - \mu)^2]} \quad -\infty < x < \infty$$

put

$$y = \frac{x - \mu}{1}$$

$$\begin{array}{l|l} x - \mu = 1 \cdot y & x = -\infty \quad | \quad x = \infty \\ dx = 1 \cdot dy & y = -\infty \quad | \quad y = \infty \end{array}$$

$$F(y) = \frac{1}{\pi [1^2 + 1^2 y^2]} dy$$

$$= \frac{1}{\pi [1 + y^2]} dy$$

$$F(y) = \frac{1}{\pi (1 + y^2)} dy$$

$$f(y) = \frac{1}{\pi (1 + y^2)} \quad -\infty < y < \infty$$

Question

Find the median of the Cauchy distribution.

Sol the median m is given by

$$F(m) = \frac{1}{2}$$

$$F(m) = \int_{-\infty}^m f(x) dx = \frac{1}{2}$$

$$\Rightarrow \int_{-\infty}^m \frac{1}{\pi (d^2 + (x - \mu)^2)} dx = \frac{1}{2}$$

$$\Rightarrow \frac{1}{\pi} \int_{-\infty}^m \frac{1}{d^2 + (x - \mu)^2} dx = \frac{1}{2}$$

we know that

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$$

$$\Rightarrow \frac{1}{\pi} \left[\frac{1}{d} \tan^{-1}\left(\frac{x - \mu}{d}\right) \right]_{-\infty}^m = \frac{1}{2}$$

$$\Rightarrow \frac{1}{\pi} \left[\tan^{-1}\left(\frac{m - \mu}{d}\right) - \tan^{-1}(-\infty) \right] = \frac{1}{2}$$

$$\Rightarrow \frac{1}{\pi} \left[\tan^{-1}\left(\frac{m - \mu}{d}\right) - (-\pi/2) \right] = \frac{1}{2}$$

$$= \frac{1}{\pi} \tan^{-1}\left(\frac{m - \mu}{d}\right) + \frac{1}{\pi} (\pi/2) = \frac{1}{2}$$

$$= \frac{1}{\pi} \tan^{-1}\left(\frac{m - \mu}{d}\right) + \frac{1}{2} = \frac{1}{2}$$

$$\Rightarrow \frac{1}{\pi} \tan^{-1} \left(\frac{m-\mu}{\sigma} \right) = 0$$

$$\frac{m-\mu}{\sigma} = 0$$

$$\boxed{m = \mu}$$

Question

Find the characteristic function of the standard Cauchy distribution.

Sol

By definition

$$\begin{aligned} \phi(t) &= E(e^{itx}) = \int_{-\infty}^{\infty} e^{itx} f(x) dx \\ &= \int_{-\infty}^{\infty} e^{itx} \cdot \frac{1}{\pi(1+x^2)} dx \\ &= \frac{1}{\pi} \int_{-\infty}^{\infty} e^{itx} \cdot \frac{1}{(1+x^2)} dx \quad \text{--- (1)} \end{aligned}$$

To evaluate, Consider Laplace distribution or double exponential distribution.

$$f(z) = \frac{1}{2} e^{-|z|} \quad -\infty < z < \infty$$

$$\phi_z(t) = E(e^{itz})$$

$$= \int_{-\infty}^{\infty} e^{itz} f(z) dz$$

$$= \int_{-\infty}^{\infty} e^{itz} \frac{1}{2} e^{-|z|} dz$$

$$= \frac{1}{2} \int_{-\infty}^{\infty} e^{itz} \cdot e^{-|z|} dz$$

(9)

using $e^{i\theta} = \cos\theta + i\sin\theta$

$$\begin{aligned} \phi_2(t) &= \frac{1}{2} \int_{-\infty}^{\infty} (\cos(tz) + i\sin(tz)) e^{-|z|} dz \\ &= \frac{1}{2} \int_{-\infty}^{\infty} \cos(tz) e^{-|z|} dz + \frac{1}{2} \int_{-\infty}^{\infty} i\sin(tz) e^{-|z|} dz \\ &\quad \text{even function} \qquad \text{odd function} \end{aligned}$$

$$\phi_2(t) = \frac{1}{2} \int_0^{\infty} \cos(tz) e^{-z} dz$$

By using

$$\int_0^{\infty} e^{-mx} \cos nx = \frac{m}{m^2 + n^2}$$

Here $m=1$ and $n=t$

$$\phi_2(t) = \frac{1}{1+t^2}$$

Since $\phi(t)$ is absolutely integrable in $(-\infty, \infty)$,

we have by uniqueness Theorem.

$$f(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ctz} \phi_2(t) dt$$

$$\frac{1}{z} e^{-|z|} = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ctz} \cdot \frac{1}{1+t^2} dt$$

Replace t by $-t$

$$e^{-|z|} = \frac{1}{\pi} \int_{-\infty}^{\infty} e^{itz} \cdot \frac{1}{1+t^2} dt$$

Replace t by z

$$\frac{1}{z} e^{-|z|} = \frac{1}{\pi} \int_{-\infty}^{\infty} e^{itz} \cdot \frac{1}{1+z^2} dz$$

interchange 'z' by 'x'

$$e^{-|t|} = \frac{1}{\pi} \int_{-\infty}^{\infty} e^{itx} \cdot \frac{1}{1+x^2} dx$$

put in ϕ_x

$$\underline{\underline{\phi_x(t) = e^{-|t|}}}$$

of 'y' is an Cauchy Variate with parameter ' μ ' and ' N ' so

$$X = \frac{Y - \mu}{N}$$

$$Y = \mu + Nx$$

$$\phi_y(t) = E(e^{ity})$$

$$= E(e^{it(\mu + Nx)})$$

$$= e^{it\mu} \cdot E(e^{itNx})$$

$$= e^{it\mu} \cdot \phi_x(Nt)$$

$$= e^{it\mu} \cdot e^{-|Nt|}$$

$$\underline{\underline{\phi_y(t) = e^{it\mu - |Nt|}}}$$